GATE TRIAGE FOR CONTAINER TERMINALS

# Rung

**Keep the gate open when the data goes bad.**

**Code**
github.com/Hratchg/rung

**Demo video**
Latest release

## The problem

Every truck leaving a container terminal goes through the gate. Five systems describe each box, and they disagree all the time.

**On a bad night the safe move is to close the gate, and every truck in line pays for it.**

**191** boxes waiting to leave Pier 7 tonight

**129** trucks booked through 6 gate lanes

## How it works

### 1 Five sources

Carrier · Terminal · Customs · Weighbridge · Reefer

Read for every box, each field badged live or simulated.

### 2 Trust and conflicts

Trust fades as a reading ages and drops 30 percent for each conflict.

Weight · Records · Reefer · Dark feed · Clock

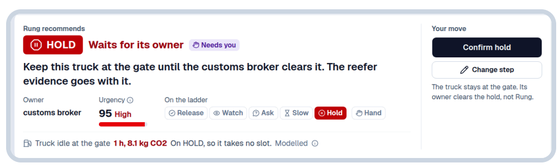
### 3 The five step ladder

| | |
|---|---|
| Hold | Waits for owner |
| Slow | New slot |
| Ask | Fresher facts |
| Watch | Recheck |
| Release | Cleared |

**Hand** — A person decides

15 small rules vote. The gentlest safe step wins.

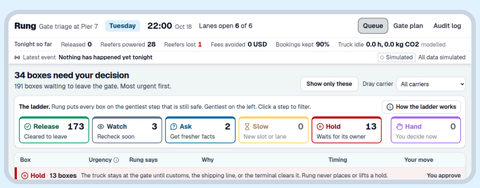
### 4 The planner approves




Every hold and hard release waits for one tap, and lands in the audit log with its reason.

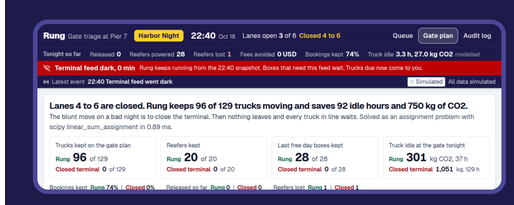
## Dual use: one app, two nights

### An ordinary Tuesday




- A dispatcher's pull list: 34 of 191 boxes need a decision.
- The voice agent calls truckers about weight mismatches.

### Harbor Night




- Half the lanes close, a customs hold lands, a beef reefer warms, the feed goes dark.
- Rung re-plans every truck in under 1 ms.

## Harbor Night results

**Lanes 4 to 6 down, feed dark. Rung against closing the terminal.**

**96** of 129
**trucks kept moving**
0 if the terminal closes

**20** of 20
**reefers kept**
0 if the terminal closes

**28** of 28
**last free day boxes kept**
0 if the terminal closes

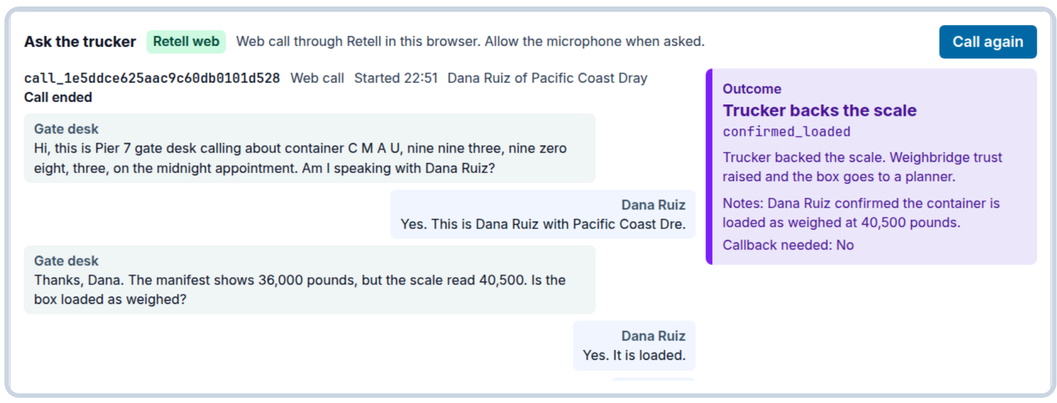
about **750** kg
**truck idle CO2 avoided**
92 idle hours, modelled

## The Retell voice agent

Manifest 36,000 lb, scale 40,500 lb. Rung asks.

1. Pier 7 gate desk, a Retell voice agent, calls the trucker.
2. The trucker backs the scale, one of four fixed outcomes.
3. The box moves to **Hand** and the planner decides.

**The model runs the call. Rung's rules decide what happens next.**




A real Retell web call, with its call id in the audit log.

## Go global

**Global by design.** The decision core never changes: one ladder, one trust model, one gate optimizer, and a person approves every hold. A region profile swaps the edges:

| | Los Angeles | Rotterdam | Shanghai |
|---|---|---|---|
| **Data** | Terminal49 | Portbase | China Single Window |
| **Units** | lb, USD | kg, EUR | kg, CNY |
| **Customs** | CBP | Douane | China Customs |

**Lagos, where trucks queue for days, needs it most.**

## Federal funding pathway

**Launch** PRIMARY — **NSF SBIR Phase I**, America's Seed Fund (NSF 26-510). Up to $305K over 6 to 18 months to prove Rung on live San Pedro Bay terminal data.

**Harden** ALTERNATIVE — **DoD SBIR** with USTRANSCOM and SDDC for strategic seaports such as Long Beach.

**Scale** ALTERNATIVE — Port-led **MARAD PIDP** grants.

**One tool for ordinary Tuesdays and Harbor Nights.**


**Team: Hratch Ghanime, George Demirjian**


Pier 7 is simulated with a fixed seed. The voice call is real.
CO2 is modelled at 8.14 kg per truck idle hour (DOE and Argonne, EPA).
Built with FastAPI, scipy, React, Retell AI, and Terminal49.